To do estimate: 180 Minutes

2026 Evil Swerve Drive To Do

File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
AgitatorSystem	add state machine to detect stall, initiate reverse, then go back to normal when done. currently 0.6 seconds to detect stall, reverse for 1.0 seconds. might consider reducing both of these a little.	very high	1	Open	1	20	
AgitatorSystem	add rate limiting on motor output so we aren't step changing the motor from forward to reverse.	very high	1	Open	1	5	
AutoSystem	add code to take care null exception that happens when user didn't select an auto.	Very High	1	Open	1	10	
IntakeSystem	CANT USE THE RATE LIMIT IN TWO PLACES. Only rate limit the final value. - remove the rate limit here... intakeAngleMotorDemand=IntakeArmRateLimit.calculate(---------------------------- IntakePositionControl.PosCtrlExec(intakeAngleTarget, IntakeArmAngleActual)); - keep the rate limit here... intakeAngleMotorDemand = MathUtil.clamp(IntakeArmRateLimit.calculate(intakeAngleMotorDemand + intakeAngleGravityConstant), tmpLowClamp, 1.0);	Very High	1	Open	1	5	
SwerveVision	Set actual camera names	Very High	1	Open	1	5	
SwerveVision	The physical camera offset position will have to be set once these are known. There is a spreadsheet in programming slack channel.	very high	1	Open	1	15	
TurretLauncher	limit switch values need to be "locked" until position is away from limit switch. (limit switch position + hysteresys value --- sign depends on which limit switch.	Very High	1	Open	1	10	
TurretLauncher	clamp the auto calculated turret setpoint. "desiredTurretAngleDegrees", using actual limit values, not -100 and 100.	Very High	1	Open	1	5	
AgitatorSystem	update motor feedforward and pid tuning	High	2	Open	1	5	
ClimbSystem	Add arm up limit switch. Read encoder. Read limit switch. Dont travel up past limit switch. Don't travel down past encoder value of zero. Add commands for go up and go down and stop. Suggest constant motor output of: suggest Up 15%, Down 40%, Off 0%	High	2	Open	1	20	
PhotonVision	Configure co-processor 1. Define and calibrate april tag cameras.	high	2	Open	1	Approx 30	May not be a programmer team task !!
PhotonVision	Configure co-processor 2. Define and calibrate april tag cameras.	high	2	Open	1	Approx 30	May not be a programmer team task !!
TeleopSystem	Add buttons for climb up and climb down. If buttons not pressed send climb stop command.	High	2	Open	1	10	
TurretLauncher	Should only use automatic setpoints if odometry is good. Otherwise set constant "safe" values.	High	2	Open	1	10	
TurretLauncher	this code: //------------------------------- if (IntakeSystem.intakeAngleTarget > 45) { TurretMotorDemand = 180; } should set the desiredTurretAngleDegrees not the motor output. It also must be above the position control. If you really want to set the motor demand based on this permissive, the demand needs to be zero.	High	2	Open	1	10	
TurretLauncher	turret motor outpt is being clamped twice.	High	2	Open	1	10	
AgitatorSystem	remove AgitatorOnRev command and all code and variables that go with it.	Medium	3	Open	1	5	
MatchSystem	init method --- The getter should probably be blue...?? locNTSend.addItemBoolean("isBlue", MatchSystem::isRed);	Medium	3	Open	1	5	
SwerveModule	may want to multiply the drive output by the cosine of the spin angle error. - spin error should always be within +/- 90 degrees after optimization. - when error is near zero, cosine is near 1.0. As error increases, value decreases to zero. This can help reduce sway when changing directions.	Medium	3	Open	1	5	

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SwerveTeleop	Do you want an aux button box ??	Medium	3	Open	1	5	
SwerveVision	When using single target results, make sure that ambiguity is within a reasonable range, say less than 0.3. If not then ignore value.	Medium	3	Open	1	x20	SAVE FOR LATER
TurretLauncher	Check that tuning constants are up to date.	Medium	3	Open	1	5	
TurretLauncher	need to update launcher speed demand calculation based on testing.	Medium	3	Open	1	5	
TurretLauncher	Need a way to set turret positon demand from joystick for testing.. Increment / Decrement value. Maybe leave this as a manual mode.	Medium	3	Open	1	x25	LEAVE FOR LATER!!!!!....
TurretLauncher	Calculate turret on target and add value to network tables. (Should this value prevent launcher from shooting ????)	Medium	3	Open	1	10	think don't need
AgitatorSystem	Determine the best motor output to use for agitation.	Tuning	4	Open	1	?	
AgitatorSystem	Determine if speed control is needed.	Tuning	4	Open	1	?	
AgitatorSystem	Determine if the agitator should have a "reverse" mode that would be used when on and not shooting. Add a new cmd method to select the new mode, and update supervisory as needed.	Tuning	4	Open	1	?	
FeederSystem	See what output works best here. Likely should do speed control and set speed setpoint to be something less than the shooter. Maybe 100 RPM less…	Tuning	4	Open	1	?	
IntakeSystem	See what motor demand works for the intake motor.	Tuning	4	Open	1	?	
IntakeSystem	Tune the intake arm position control	Tuning	4	Open	1	?	
SwerveModule	Tune feedforward. Calculate maximum speed.	Tuning	4	Open	1	?	
SwerveModule	Tune PID. Current tuning is Kp 0.5, Ki 0, Kd 0 – all times Kn. Suggest if using integration, set a range, and max integral values to minimize windup. However ramp will likely require non zero Ki.	Tuning	4	Open	1	?	
AutoFunctions	Add AutoSpin function.	Low	5	Open	1	?	May not need to use. Nice to have in list of things robot can do. After GOS
AutoFunctions	Add AutoStraight function	Low	5	Open	1	?	May not need to use. Nice to have in list of things robot can do. After GOS
AutoStep	Add command for "Defense". May not need.	Low	5	Open	1		
AutoSystem	In method "readFiles", variable "fileInstance" does not have an extension so it doesn't need to be removed. This doesn't cause any issues, it just isn't needed.	Low	5	Open	1	?	After GOS
AutoSystem	Add name of selected auto to Network Tables so drivers can verify that is being used.	Low	5	Open	1	?	After GOS
Climb	Do the TODO… We don't know what climb is yet.	Low	5	Open	1	?	Need to have build define what this is….. After GOS.
IntakeSystem	This appears to be redundant.	Low	5	Open	1	?	After GOS
MatchSystem	If you want to diagnose, add a network table entry for red or blue or whatever…	Low	5	Open	1	x10	isBlue calls isRed
SupervisorySystem	climbing -- it seems like climbing is not going to be for auto -- check with build -- - suggest leaving command alone for now. If going to be used for auto likely a do climb state machine will be needed -- or something to sequence the actions…	Low	5	Open	1		
SupervisorySystem	Add a class variable for locNTsend for NT variable. Add a public static void init() function to init the network table stuff. In this create the network table helper object. Add a string item that is Mode (or something like that). Add a class level private static String locMode variable = "Initial". In each supervisory function, set the value of locMode to the name of the function that is being called. Also add a triggerupdate to each supervisory function. Add a public static String getMode() function.	Low	5	Open	1	?	
SwerveOdometry	If you have null exceptions when getting values, revist the getters to check for a null value of the position object.	Low	5	Open	1		
SwerveTeleop	Add a driver button to execute a trajectory while button is held done. When first pressed (edge triggered) set init variable in call to follow trajectory. Will need to select trajectory based on RED/BLUE (Or just for testing….)	low	5	Open	1	x20	SAVE FOR LATER

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File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
SwerveTeleop	Remove old commented out things that will never be used.	Low	5	Open	1	?	After GOS
SwerveTeleop	Add driver button box ???	Low	5	Open	1	?	After GOS
SwerveTeleop	Add aux button box ???	Low	5	Open	1	?	After GOS
AgitatorSystem	Remove unused imports	Very Low	6	Open	1	?	
AgitatorSystem	Add JavaDoc	Very Low	6	Open	1	?	Mostly done. Need description for class.
AutoFunctions	Add JavaDoc	Very Low	6	Open	1	?	
AutoRoutine	Add JavaDoc	Very Low	6	Open	1	?	
AutoStep	Suggest adding the following Cmd -- ExecuteCmd (command name is in Function string variable)	Very Low	6	Open	1	?	
AutoStep	Suggest adding the following Cmd -- ExecuteCheckFor (command name is in function string variable)	Very Low	6	Open	1	?	
AutoStep	Suggest adding the following Cmd -- AutoDone -- This will do nothing forever....similar to wait but with a forever timeout...	Very Low	6	Open	1	?	
AutoStep	Add JavaDoc	Very Low	6	Open	1	?	
AutoSystem	Add JavaDoc	Very Low	6	Open	1	?	Mostly done. Need description for class.
FeederSystem	Remove unused imports.	Very Low	6	Open	1	?	
FeederSystem	Add JavaDoc	Very Low	6	Open	1	?	
IntakeSystem	Say which motor is arm and which is intake.... Is 1 intake, and 2 arm ??	Very Low	6	Open	1	?	
IntakeSystem	Add JavaDoc	Very Low	6	Open	1	?	
IntakeSystem	Is this intake or arm motor ????	Very Low	6	Open	1	?	
IntakeSystem	what is this for? Maybe check the actual angle -- encoderRot -- instead.	Very Low	6	Open	1	?	
SwerveDrive	Add JavaDoc	Very Low	6	Open	1	?	
SwerveDrive	Remove unused imports	Very Low	6	Open	1	?	
SwerveModule	Review motor configuration. Probably leave alone.	Very Low	6	Open	1	?	
SwerveModule	Add JavaDoc	Very Low	6	Open	1	?	
SwerveOdometry	Add JavaDoc	Very Low	6	Open	1	?	
SwerveOdometry	put some comments here. For example, this is the initial robot position (pose)	Very Low	6	Open	1	?	
SwerveTeleop	JavaDoc	Very Low	6	Open	1	?	
SwerveVision	JavaDoc	Very Low	6	Open	1	?	
TrajectorySystem	Add JavaDoc	Very Low	6	Open	1	?	
AutoSystem	Read of autos does NOT work. The code that tries to open the file fails. Two reasons: 1) Directory of AUTO_DIR is not used as parent, 2) File extension, AUTO_FILE_EXTENSION, is not added to name.	Very High	1	Complete	3		Error is caught by Try/Catch block, but no autos are read.
AutoSystem	Variable autoHashMap was not being created. Code was crashing with null exception. Add variable initialization in init function.	Very High	1	Complete	3		
AutoSystem	All "getters" that use variable "locExecRoutine" will break when the value of this variable is null. Need to check for not null "if (locExecRoutine != null) {" and only get the value when this is true. Otherwise return a default value, like 0.0.	Very High	1	Complete	3		This causes the entire code to crash!!
AutoSystem	Does not compile. Replace with generic Exception instead of FileNotFoundException.	Very High	1	Complete	3		
SwerveModule	The NEO Vortex is NOT use spark max. I think it uses a Spark Flex controller. These are different. Need to read and convert.	Very High	1	Complete	3		
TurretLauncher	The calculate of turret angle and distance isn't correct. The geometry methods don't update themselves they return new objects. So Line 149 - TurretOffset.rotateBy(new Rotation2d(SwerveOdometry.getrotposition())); Needs to be TurretOffset = TurretOffset.rotateBy(new Rotation2d(SwerveOdometry.getrotposition())); Same with line 150.	Very High	1	Complete	3		
TurretLauncher	The variblesdesiredTurretAngleDegrees needs to be clamped to be within the limits of the turret. For now lets use -100.0, 100.0	Very High	1	Complete	3		

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File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
TurretLauncher	Add speed control for launcher. One PID will create output value for both motors. Add motors, Add sparkmax encoders, create average velocity, add simple motor feedforward and PID. (see swervemodule drive controller.)	Very High	1	Complete	3		Partially done. Sensors not read. Both motor velocity not averaged. Encoder object not defined in init. No need to negate. Negate is done in motor config…
TurretLauncher	Convert to degrees. Think that the default is rotations. Also take into account gear ratio...	Very High	1	Complete	3		Gear ratio is 200/20. Gear box is 16/1. Will be 4/1 soon. So for now it is 160/1, but will be 40/1.
AutoRoutine	set class variable with value of autoDescription parameter..	High	2	Complete	3		
AutoRoutine	set class variable with value of autoDescription parameter.. (2nd constructor)	High	2	Complete	3		
AutoSystem	In method "readFiles", parsing of the variable "Funcname", leaves leading and trailing spaces and does not remove surrounding double quotes. Suggest using .trim function to first remove leading and trailing white space. Then if the first and last characters are both ", then remove them.	High	2	Complete	3	x5	Only partially done. The leading and trailing white space has been removed
AutoSystem	Add "ExecuteListInit" to Robot.java. Add "ExecuteList" to robot.java.	High	2	Complete	3		
AutoSystem	Add code to the init function to get all available autos, then call the function to read them into the auto cache…	High	2	Complete	3		
AutoSystem	Add private static HashMap to store all trajectories <String,Trajectory<SwerveSample>	High	2	Complete	3		
AutoSystem	Add your newRoutine to the arrayList routines...	High	2	Complete	3		
AutoSystem	Add your newRoutine to the class variable to be created containing the HashMap	High	2	Complete	3		
AutoSystem	If you store the routines in the class variable containing the HashMap, then this routine doesn't need to return anything (void)	High	2	Complete	3		
AutoSystem	Add a function to return an AutoRoutine from the hashmap with a calling String parameter of the name (file name without extension)	High	2	Complete	3		
AutoSystem	Add call to TrajectorySystem.FollowTrajectory	High	2	Complete	3		Needed to add this to the individual case….
IntakeSystem	scale arm position.. default is rotations.... so DEG = ROT * 360 / GEAR_RATIO. Since encoder position will start at 0 when booted, add 90.0 to the equation.	High	2	Complete	3		
IntakeSystem	Use built in spark max encoder. We are no longer getting a through bore encoder..	High	2	Complete	3		
IntakeSystem	Encoder definition isnt correct.. Now use spark max built in encoder .GetEncoder(); see above.	High	2	Complete	3		
IntakeSystem	limit switch is new wired to roboRIO... so this won't work. Use digital input.. Need to define object (class level), create object(in init function), read object (here).	High	2	Complete	3		
TrajectorySystem	Make this modulo +/- PI... There is a WPILIB function for this... MathUtil.angleModulus	High	2	Complete	3	x5	Sorry. This wasn't clear. Only the error portion of this needed the angle modulus.
TrajectorySystem	add calculate of "complete" and return "complete" to user. complete = elapsedtime >= trajectory time length AND onTarget.	High	2	Complete	3		
TurretLauncher	Need a calculation of " launchertargetSpeed" For now it could be as simple as: launchertargetSpeed = 100.0; as a placeholder. Or it could be" launchertargetSpeed = 1000.0 + TurretDistance * 200.0;	High	2	Complete	3		
TurretLauncher	Line 179 - launchertargetSpeed = 0.0; should be: useSpeedTarget = 0.0;	High	2	Complete	3		
TurretLauncher	Remove turret motor config. Config burned into motor from Rev Hardware Client..	High	2	Complete	3		added comments
TurretLauncher	"readSensors" is never called. Also object should be defined in init, not during each execution. Read both launcher encoders and average values. It is okay to leave launch encoder velocity in RPM.	High	2	Complete	3		
TurretLauncher	More definition of turret position control object to init from excute	High	2	Complete	3		
TurretLauncher	get the values of the goal position based on alliance. I think there is a spreadsheet in slack. For now just use RED target.	High	2	Complete	3		

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File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
TurretLauncher	This will not be throughbore encoder any longer. Use built in SparkMax encoder....	High	2	Complete	3		
TurretLauncher	Use spark max built in encoder instead. Build didn't mount through bore encoder....	High	2	Complete	3		
TurretLauncher	getPosition can be used above....	High	2	Complete	3		
AgitatorSystem	If there is any motor configuration, comment out. Use Rev Hardware client to set all motor configurations.	Medium	3	Complete	3	x5	
AgitatorSystem	Set motor CAN Id	Medium	3	Complete	3		
AutoFunctions	Add new java file for static Auto processing functions.	Medium	3	Complete	3		
AutoFunctions	Add AutoFunctions.java file for static functions. Add an boolean AutoWait( ) function. It should set drive demand to zero and return false..	Medium	3	Complete	3		
AutoSystem	Add Network table variable to display the elapsed time for each step. Suggest adding class variable private static double locStepTime = 0.0;. Then add this item to network tables locNTSend.addItemDouble(“StepTime”, AutoSystem::getStepTime);. Then set the value of this variable in ExecuteList right before checking if there is a non-zero timeout value. LocStepTime = autoTimer.get(); Then add a getter method public static double getStepTime() and return locStepTime.	Medium	3	Complete	3		
AutoSystem	Rename method availableTrajectories to availableAutos and change references….	Medium	3	Complete	3		
AutoSystem	Update Robot.java to read selected auto from network tables and pass to our auto routines..	Medium	3	Complete	3		
AutoSystem	Publish list of read autos to Network Tables.	Medium	3	Complete	3		
AutoSystem	Add items to NT tables for debugging and watching autos. Suggest Step, Timeout, Parm1, Parm2, Parm3, Function. Add public getters for these.	Medium	3	Complete	3		This may be done
AutoSystem	add locNTSend trigger update.	Medium	3	Complete	3		
AutoSystem	Add call to AutoFunctions.AutoWait().	Medium	3	Complete	3		
FeederSystem	If any motor configuration exists, comment out. Use Rev Hardware client to set motor configuration.	Medium	3	Complete	3		LEAVE FOR LATER!!!!!….
FeederSystem	Set feeder motor CAN ID	Medium	3	Complete	3		
IntakeSystem	Comment out the motor configuration information. Configuration can be set by Rev Hardware Client.	Medium	3	Complete	3		LEAVE FOR LATER!!!!!….
MatchSystem	Make setRed and SetBlue methods private. The execDisabled should be the only thing that calls this.	Medium	3	Complete	3	x2	
MatchSystem	disableExec will be called by robot.java. It won’t use the result, so make this function void.	Medium	3	Complete	3	x2	
MatchSystem	There are “DriverStation” functions that can be called to get match information. Specifically there is a getAlliance function. Sadly it returns an Optional. So you first have to check for .isPresent, then test if their alliance is UNKNOWN, RED, or BLUE.	Medium	3	Complete	3	x10	
MatchSystem	Suggest adding a “MatchSystem.java” file. This would be a pseudo static class like the other systems. This would have internal variables indicating if 4150 was Red or Blue. Add a method “disableExec” to be called when disabled to get the match info. Add “getter”functions to see if we are red or blue.	Medium	3	Complete	3	x15	
SwerveDrive	Add actual positions of modules.	Medium	3	Complete	3		May need to measure first..
SwerveDrive	Change max speed from 13.0 to 15.0 This is a guess. This is done because the NEO Vortex motors are more powerful and have a higher maximum no load speed.	Medium	3	Complete	3		
SwerveModule	Change max speed from 13.0 to 15.0 feet/sec. This is 4.572 meters/sec (approx). This is a guess. Change Drive_Kn = 1/4.572. (Approx line 38)	Medium	3	Complete	3		

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File	Todo	Priority	Priority Numb	Status	Status Numb	Estimated Complete Time (Minutes)	Notes
SwerveModule	Measured circumference of swerve wheels is 32 centimeters, or 12.5984 inches. This translates to a diameter of 4.01019527 inches. Robot currently usede 3.705. Compute a new driveDistFactor and update code. (Approx line 88)	Medium	3	Complete	3		
SwerveTeleop	Add a network table variable for Field oriented driving. locNTsend.addItemBoolean("FieldOriented", SwerveTeleop::getFieldOrientedDriving);	Medium	3	Complete	3		
TrajectorySystem	Set the values of yErr and rotErr. Take the code from the calculation of yvelDmd and rotVelDmd. This will be almost identical to what was done with xErr.	Medium	3	Complete	3		
TrajectorySystem	Set class variable elapsedTrajTime = elapsedTime in followtrajectory method.	Medium	3	Complete	3		
TrajectorySystem	set class variable timeLengthOfTrajectory in followtrajectory in the “init” if block of code.timeLengthOfTrajectory = TrajectoryToRun.getTotalTime() right after setting TrajectoryToRUn =	Medium	3	Complete	3		
TrajectorySystem	set class variable done in followtrajectory. Done = complete. Right after complete is calculated.	Medium	3	Complete	3		
TrajectorySystem	call locNTsend.triggerUpdate() at at end of followtrajectory method	Medium	3	Complete	3		
TrajectorySystem	add network table entries for ontarget, done, havesample, timelengthoftrajectory (rename if you don't like name)	Medium	3	Complete	3		Some of this may be done.
TrajectorySystem	TODO: add getters for new things written to network tables. ontarget, havesample,done.	Medium	3	Complete	3		This may be duplicate….
TurretLauncher	Set “dump to our side” field positions Red and Blue (also maybe left and right)	Medium	3	Complete	3	x5	
TurretLauncher	Use MatchSystem getter to determine red or blue field side. Need to pick alliance dump location and target location based on red/blue.	Medium	3	Complete	3	x10	Partially complete. Added more in description.
TurretLauncher	Need to use the limit switches to stop travel. When we get a limit siwtch EDGE set the motor control position. (Make certain to back calculate rotations from degrees.)	Medium	3	Complete	3	x15	
TurretLauncher	Add LauncherSpeedDemand to network tables	Medium	3	Complete	3		
TurretLauncher	Calculate launcher speed on target (within 75 RPM) and publish to network tables.	Medium	3	Complete	3		
TurretLauncher	Set physical offset of robot to turret	Medium	3	Complete	3		Need measurement – it is approximately 6.75 from edge. Center of robot is 12.5. so offset is 5.75 Inches or 0.14605 meters
TurretLauncher	Set on / off command for launcher.	Medium	3	Complete	3		
TurretLauncher	Add limit switch values to network tables.	Medium	3	Complete	3		
TurretLauncher	TODO: Set CAN ID	Medium	3	Complete	3		
AutoRoutine	Remove unused imports	Low	5	Complete	3		
AutoSystem	Set variable AUTO_FILE_EXTENSION as final	Low	5	Complete	3	?	
FeederSystem	Might need to do speed control here... Copy code from launcher.	Low	5	Complete	3	?	After GOS ???
IntakeSystem	determine up and down angles on the robot -- suggest starting with 90 is up and 0 is down...	Low	5	Complete	3		
IntakeSystem	When low limit switch EDGE is true, set position to known value and send this to motor controller. Remember to reverse the scaling.	Low	5	Complete	3		
SupervisorySystem	Do you want a command to play defense? (Or is there one already) This would turn everything off and retract the intake to 90 degrees.	Low	5	Complete	3	?	
TurretLauncher	Remove unused imports	Low	5	Complete	3		
SupervisorySystem	Add a private constructor.	Very Low	6	Complete	3	?	
SwerveModule	Remove unused imports	Very Low	6	Complete	3		
TrajectorySystem	Remove unused imports.	Very Low	6	Complete	3	?	

Name	Priority
High	2
Low	5
Medium	3
Tuning	4
Very High	1
Very Low	6

Status	Priority
Open	1
Partially Complete	2
Complete	3

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